Royal Crest Academy
BIKITA · MASVINGO

O LEVEL GEOGRAPHY · WEATHER & CLIMATE

Three ways *the sky makes rain.*



All rain follows one rule. **What pushes the air up gives each type its name.**

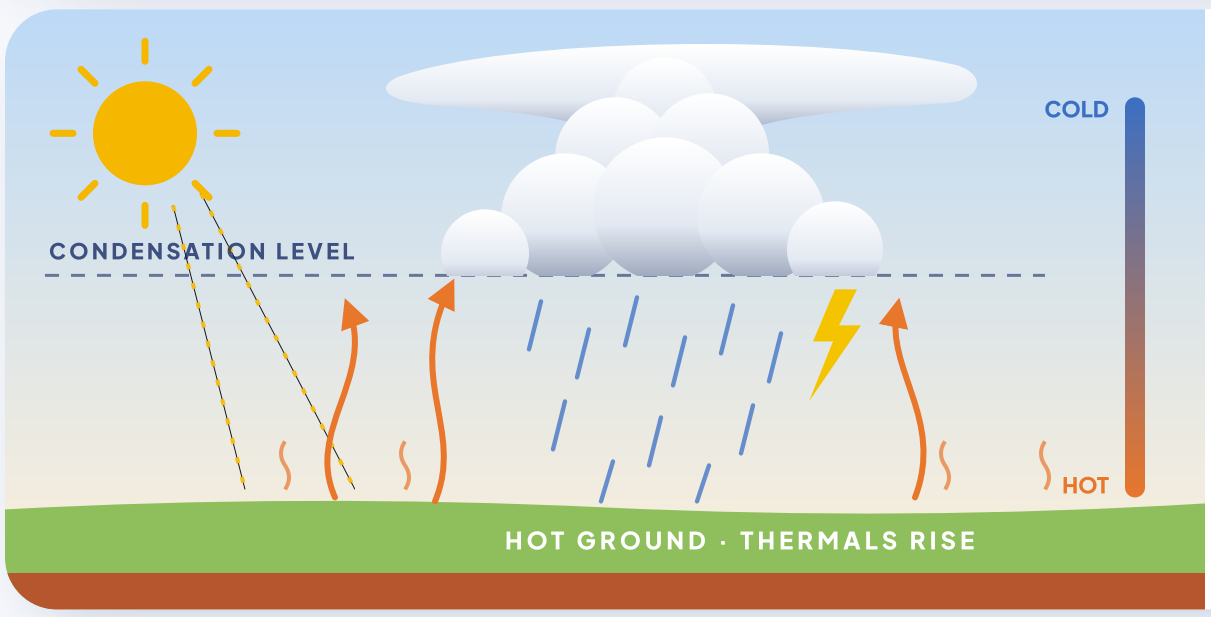
THE ONE RULE BEHIND ALL RAIN

Air rises — pushed up by heat, hills or a front

> **It cools** — about 1 °C for every 100 m

> **Cloud forms** — vapour condenses at dew point

> **Rain falls** — droplets grow heavy enough



1 Convectional rain

The sun pushes it up.

- The sun heats the ground. The air above warms, expands and rises in thermals.
- It cools to dew point and condenses into cumulus clouds.
- Strong uplift builds towering cumulonimbus.

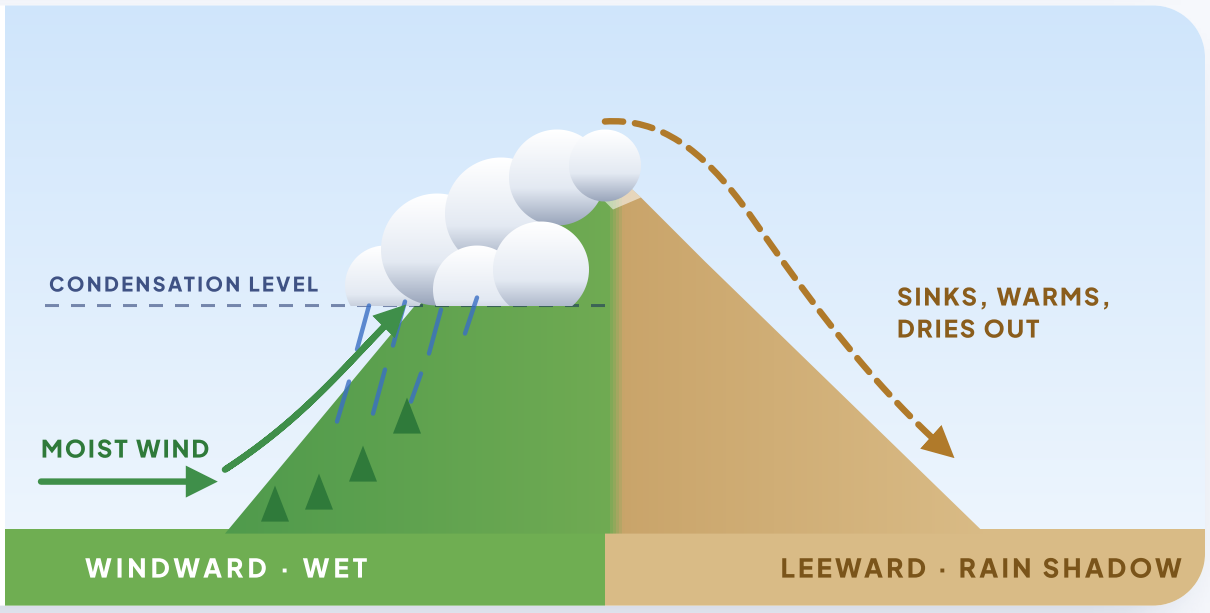
CLOUD Cumulonimbus
RAIN **Heavy, short afternoon storms** · thunder and lightning
WHERE **Zimbabwe, November to March** · most of our rain

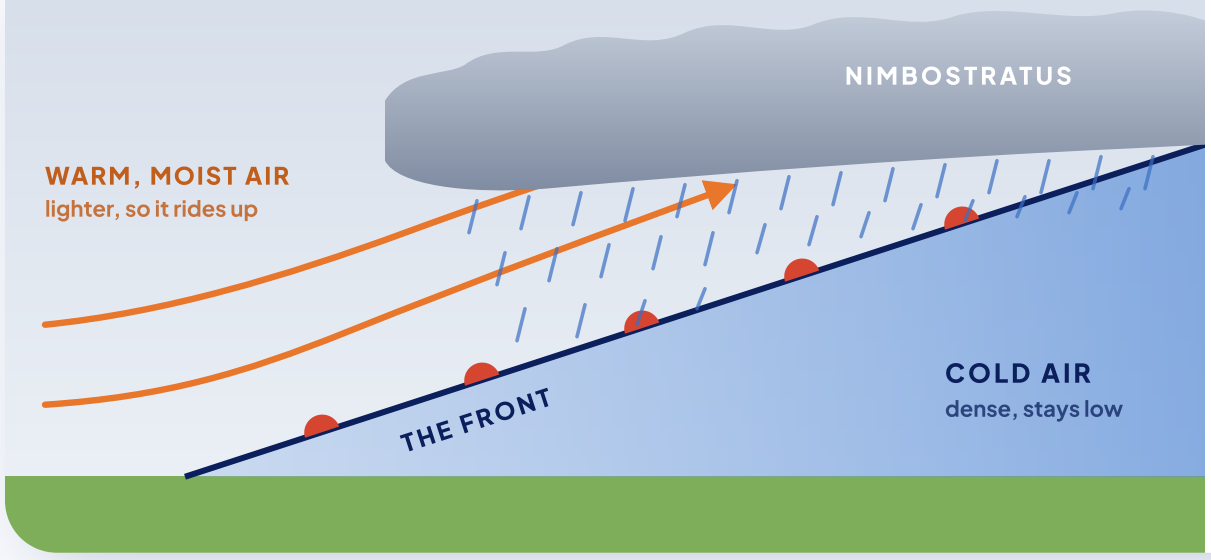
2 Relief rain

A mountain pushes it up.

- Moist wind is forced up the windward slope of high ground.
- It cools, condenses and rains on the windward side.
- Beyond the top the air sinks, warms and dries: the rain shadow.

CLOUD **Cloud caps on the hills**
RAIN **Wet windward slopes** · dry leeward side
WHERE **Eastern Highlands: Nyanga, Chimanimani** · the Save valley lies in the rain shadow





3 Frontal rain

Cold air pushes it up.

- Warm, moist air meets a mass of cold, dense air at a front.
- The lighter warm air is forced to rise over the cold air.
- It cools slowly into wide grey sheets of nimbostratus.

CLOUD Nimbostratus
RAIN **Steady, light to moderate** · lasts for hours
WHERE **Most of Britain's rain** · in Zimbabwe, winter cold fronts bring cool, grey days

EXAM TIP · ZIMSEC & CAMBRIDGE Draw and label it: arrows for moving air, a dashed condensation level, the cloud's name and where the rain falls.

Royal Crest Academy
SCIENCE · INNOVATION · SKILL

We read the sky every morning at our own weather station.

Form 1 · 2027 enrolment open
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